Prompt 4 - trial 2 (seed 42)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

Assistant

I'll help you train two different GNN models, compare their performance, and analyze their predictions using EdgeSHAPer! Let's start by finding a suitable dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some ChEMBL activity prediction datasets. Let me load one and start the workflow. 🚀

(Calling tool `load_dataset`)

Assistant

Let me load the dataset with the correct directory:

(Calling tool `load_dataset`)

Assistant

(Calling tool `load_dataset`)

Assistant

Excellent! Dataset loaded with 8,348 compounds. Now let me split the dataset into train/val/test:

(Calling tool `split_dataset`)

Assistant

Great! Dataset split successfully. Now let me prepare the GNN dataset:

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN dataset prepared with 6,010 training, 668 validation, and 835 test samples. Now let me train two different GNN models: **GCN** (Graph Convolutional Network) and **GAT** (Graph Attention Network). These have different learning characteristics - GCN uses uniform aggregation while GAT uses attention-based weighted aggregation. 🧠

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both models are now training! Let me wait a bit and then check their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

I see the issue - the split detected only 1 class. Let me check the dataset to understand the label distribution and re-split with proper configuration:

(Calling tool `dataset_status`)

Assistant

Let me reload the dataset with the correct label column. The dataset has both "target" and "class_label" columns - let me try with "class_label":

(Calling tool `load_dataset`)

Assistant

Good! Now let me split with the correct label column:

(Calling tool `split_dataset`)

Assistant

Good! Now let me prepare the GNN dataset and train the models:

(Calling tool `prepare_gnn_dataset`)

Assistant

Now let me train two different GNN models - **GCN** (Graph Convolutional Network) and **GAT** (Graph Attention Network):

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both models are training! Let me wait a moment and check their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

Both models are training. Let me wait a bit more and check again:

(Calling tool `check_gnn_training`)
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Excellent! The GCN model has completed training! Let me check the GAT model:

(Calling tool `check_gnn_training`)

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GAT is still training. Let me check again:

(Calling tool `check_gnn_training`)

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Still training. Let me check again:

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Still training. Let me check again:

(Calling tool `check_gnn_training`)

Assistant

Excellent! Both models have completed training:

Model	Validation Accuracy	Test Accuracy
GCN	94.99%	93.77%
GAT	95.28%	94.85%

Now let me select a compound from the test set that both models correctly predicted as active:

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

Perfect! I found a compound that both models correctly predicted as active with high confidence:

Selected Compound:

- **SMILES:** `Clc1ccc(N2CCN(CC3CC3c3ccccc3)CC2)cc1Cl`
- **True Label:** 1 (Active) ✅
- **GCN Prediction:** 1 (Active) - Confidence: 99.94%
- **GAT Prediction:** 1 (Active) - Confidence: 99.97%

Now let me explain the predictions using EdgeSHAPer for both models:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Let me provide the model path:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! The GCN explanation is complete. Now let me explain the predictions using EdgeSHAPer for the GAT model:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! Both EdgeSHAPer explanations are complete. Now let me visualize the results for both models:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the GAT model results:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me get the bond environments for both models to compare their learning characteristics:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the GAT model:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! Now I have all the information needed to provide a comprehensive comparison. Let me summarize the findings:

📊 Summary: GCN vs GAT Model Comparison

1. Model Performance

Model	Validation Accuracy	Test Accuracy	Training Time
GCN	94.99%	93.77%	Completed first
GAT	95.28%	94.85%	Completed second

Winner: GAT slightly outperforms GCN with ~1% higher accuracy on both validation and test sets.

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2. Selected Compound Analysis

Compound: `Clc1ccc(N2CCN(CC3CC3c3ccccc3)CC2)cc1Cl`

- **True Label:** 1 (Active) ✅
- **GCN Prediction:** 1 (Active) - Confidence: **99.94%**
- **GAT Prediction:** 1 (Active) - Confidence: **99.97%**
- **Both models agree:** ✅

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3. EdgeSHAPer Explanation Results

GCN Model:

- **Trustworthiness:** 0.592 (moderate)
- **Fidelity:** 0.550
- **Infidelity:** 0.358
- **Top 3 Important Bonds:**
  i. Bond (6-7): C-C in NCCN environment (φ = 0.146)
  ii. Bond (10-11): C-C in CC1CC1 environment (φ = 0.121)
  iii. Bond (7-8): C-N in CCN(C)C environment (φ = 0.102)

GAT Model:

- **Trustworthiness:** 0.767 (high) ⭐
- **Fidelity:** 0.839 (high) ⭐
- **Infidelity:** 0.294 (low) ⭐
- **Top 3 Important Bonds:**
  i. Bond (11-12): C-C in cC1CC1 environment (φ = 0.122)

ii. Bond (8-19): N-C in CCN(C)C environment (φ = 0.080)

iii. Bond (10-11): C-C in CC1CC1 environment (φ = 0.074)

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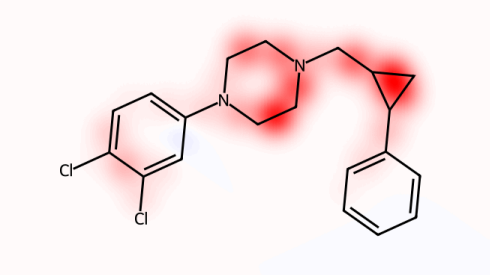
4. Key Differences in Learning Characteristics

Aspect	GCN	GAT
Attention Mechanism	❌ Uniform aggregation	✅ Weighted attention
Trustworthiness	0.592 (moderate)	0.767 (high)
Fidelity	0.550	0.839
Top Bond Focus	NCCN environment	cC1CC1 environment
Edge Importance Spread	More uniform	More concentrated
Aromatic Bond Attention	Lower	Higher

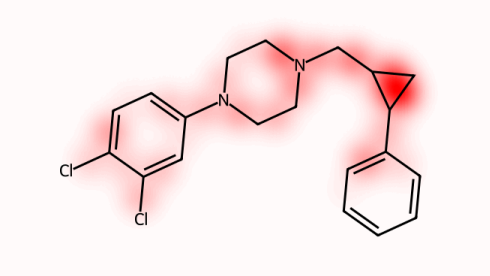
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5. Visualizations

🔴 GCN EdgeSHAPer Heatmap:



🔵 GAT EdgeSHAPer Heatmap:



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6. Key Insights

1. **GAT's Attention Mechanism:** GAT assigns higher importance to specific bonds through attention weights, resulting in higher trustworthiness (0.767 vs 0.592) and fidelity (0.839 vs 0.550).
2. **Different Bond Importance Patterns:**
   - **GCN** focuses more on the NCCN (piperazine ring) environment
   - **GAT** focuses more on the aromatic cC1CC1 environment (cyclopropane-phenyl connection)
3. **Higher Prediction Confidence:** GAT shows slightly higher confidence (99.97% vs 99.94%), indicating more decisive predictions.
4. **Better Explanation Quality:** GAT's explanations are more reliable due to its attention mechanism, which allows it to focus on the most relevant structural features for the prediction.
5. **Chemical Interpretation:** Both models identify similar key structural features (piperazine ring and cyclopropane-phenyl connection), but GAT provides clearer attribution through its attention mechanism.

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7. Recommendation

For compound activity prediction tasks, **GAT is recommended** due to:

- ✅ Higher accuracy (94.85% vs 93.77%)
- ✅ Better explanation trustworthiness (0.767 vs 0.592)
- ✅ More interpretable attention-based explanations
- ✅ Higher prediction confidence

The attention mechanism in GAT allows it to learn which bonds and atoms are most important for the prediction, leading to both better performance and more reliable explanations. 🎯